

Practical no. 17

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Class :- CO3K – B

Subject :- DSU

Program :-

```
#include<stdio.h>
#include<stdlib.h>
#include<conio.h>
struct node{
int data;
struct node*next;
};
struct node* top=NULL;
void push(int value){
struct node* newnode=(struct node*)malloc(sizeof(struct node));
if(newnode==NULL){
printf("\nStack overflow");
return;
}
newnode->data=value;
newnode->next=top;
top=newnode;
printf("\n%d is pushed in stack.",value);
void pop(){
struct node* temp=top;
if(top==NULL){
printf("nStack underflow");
return;
}
printf("\n%d is popped from stack",top->data);
top=top->next;
free(temp);
}
void display(){
struct node* temp=top;
if(top==NULL){
printf("\nStack empty");
return;
}
printf("\nStack Elements are :- ");
while(temp!=NULL){
printf("\n%d",temp->data);
temp=temp->data;
```

```

}
printf("\n");
}
void main(){
int value,choice;
clrscr();
while(1){
printf("\n---STACK MENU---");
printf("\n1.Push\n2.Pop\n3.Display\n4.Exit");
printf("\nEnter your choice :- ");
scanf("%d",&choice);
switch(choice){
case 1:
printf("\nEnter value to be pushed :- ");
scanf("%d",&value);
push(value);
break;
case 2:
pop();
break;
case 3:
display();
break;
case 4:
return 0;
default:
printf("\nInvalid Choice!");
}
}
getch();
return 0;
}

```

OUTPUT :-

DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip

```
---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 1

Enter value to be pushed :- 10

10 is pushed in stack.
```

```
---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 1

Enter value to be pushed :- 20

20 is pushed in stack.
```

```
---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 3

Stack Elements are :-
20
10
```

```
---STACK MENU---
1.Push
2.Pop
3.Display
4.Exit
Enter your choice :- 2

20 is popped from stack
```